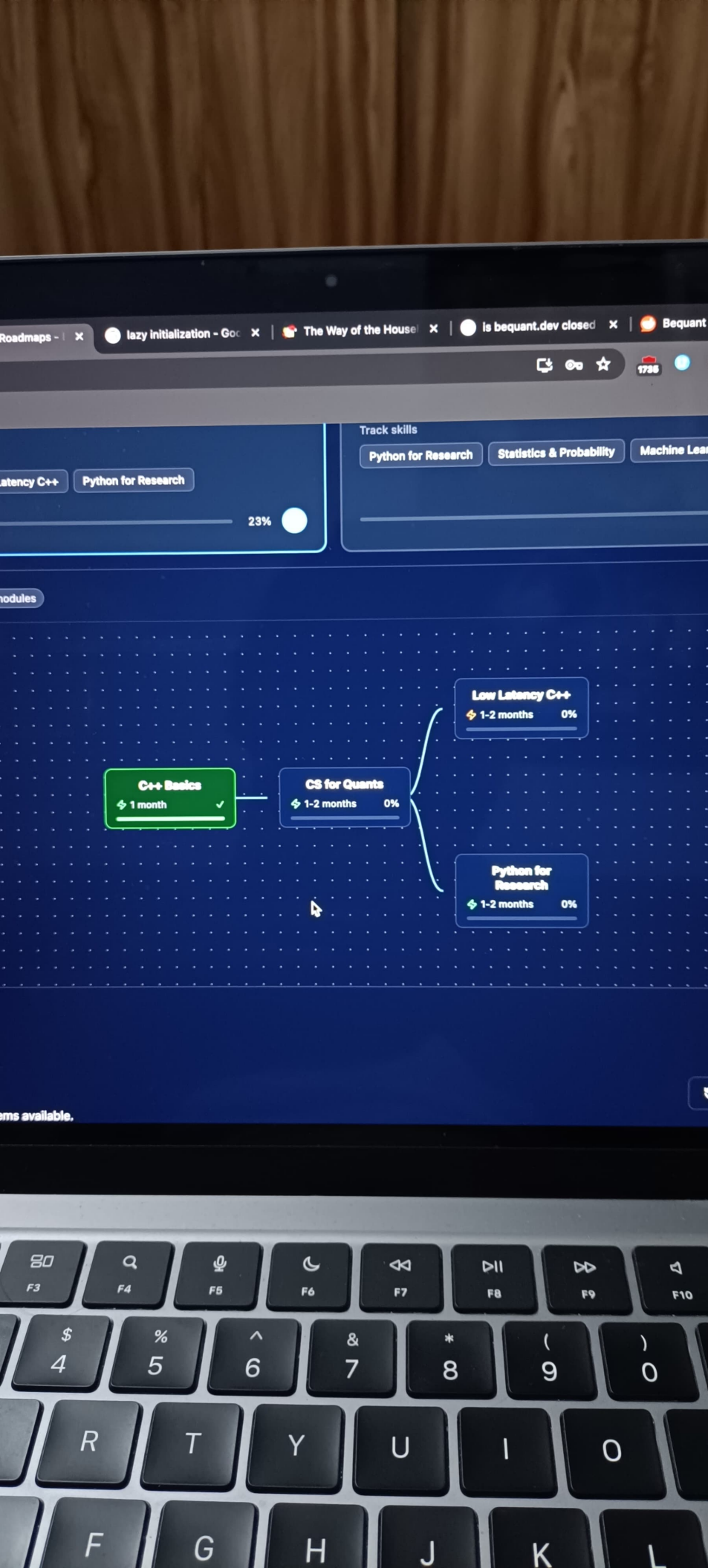
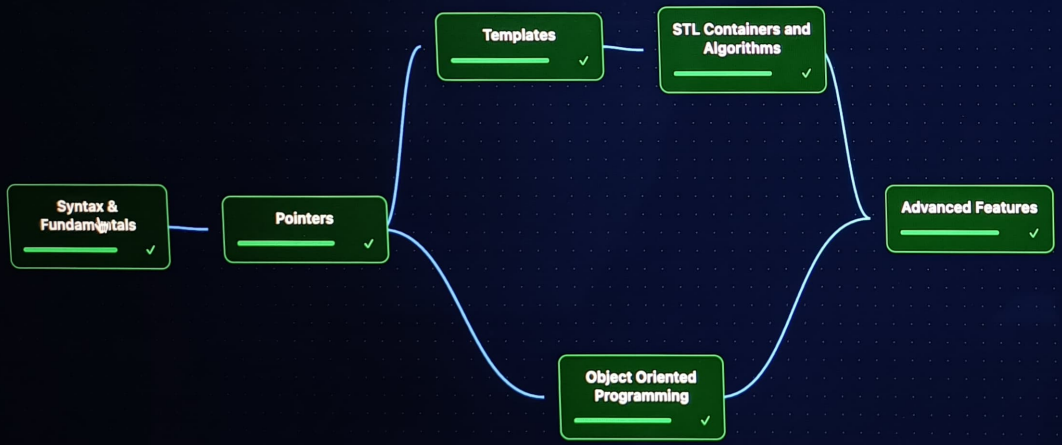

Roadmaps -
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C++ Basics roadmap
33/33 topics completed
Expected level



Topics

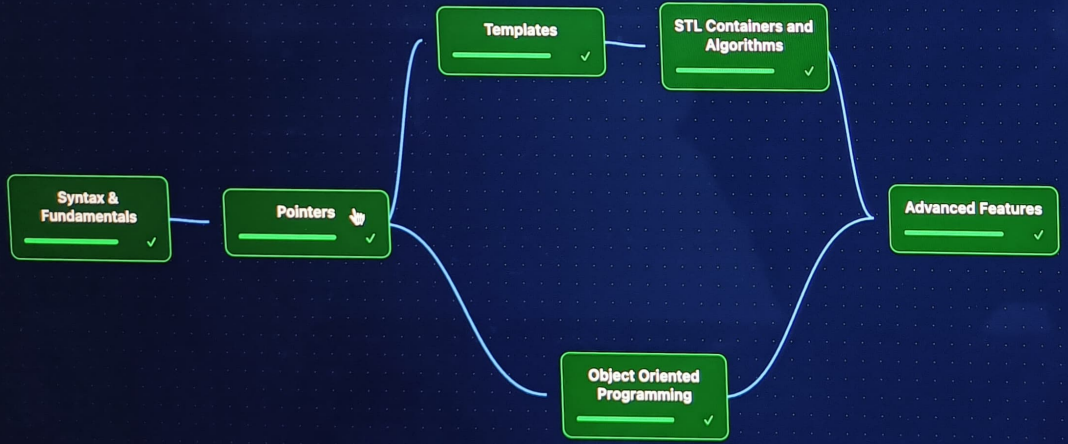
← Roadmap

- The C++ Compiler — Easy
- Types and Initialization — Easy
- Expressions and Casting — Easy
- Enums and Strong Enums — Easy
- Storage and Attributes — Medium

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C++ Basics roadmap
33/33 topics completed

Expected level



Topics

← Roadmap

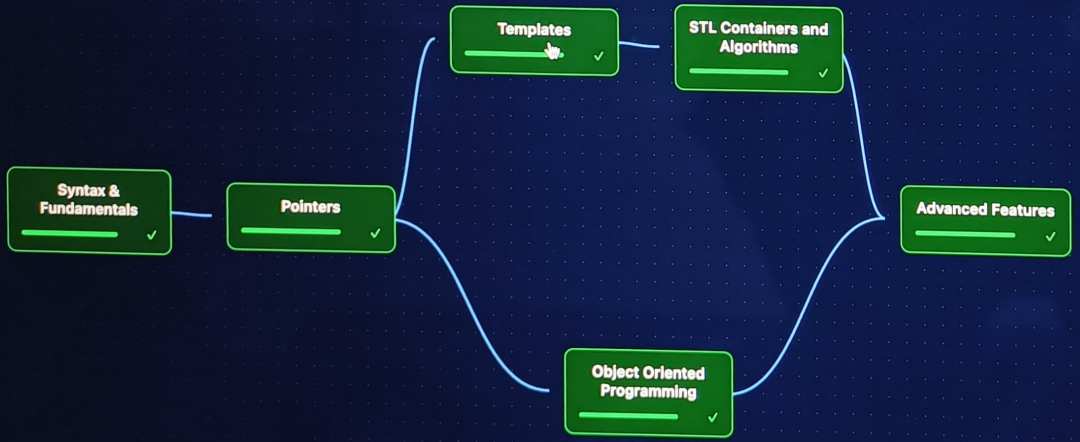
- Pointers and Free Store — Easy
- Pointers & Free Store II — Easy
- Array Copying with Pointers — Easy
- Array Reversal with Pointers — Easy
- Const and Pointers — Medium
- Function Pointers, Functors, ... — Medium

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C++ Basics roadmap
33/33 topics completed

Expected level



Topics

← Roadmap

- Type Deduction — Easy
- constexpr functions — Easy
- Introduction to Templates — Easy
- Template Specialization — Medium
- Lambdas — Medium

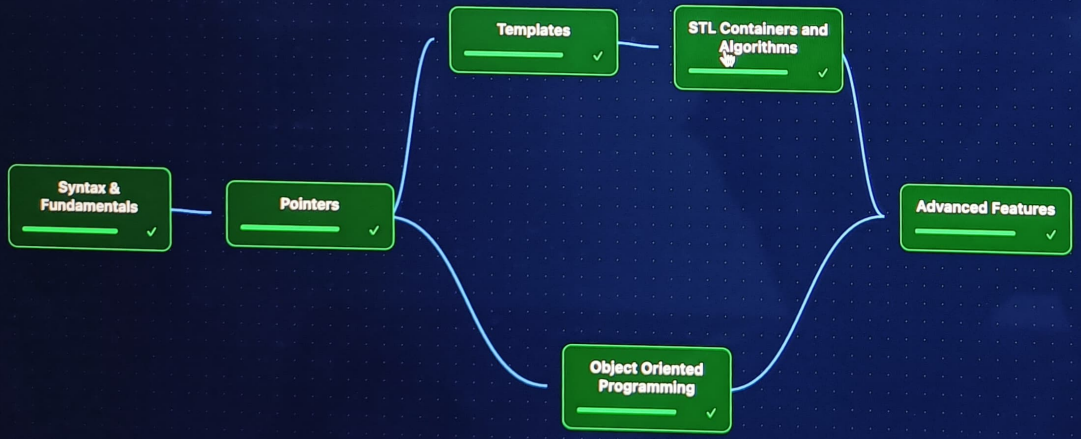
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33/33 topics completed

Expected level



Topics

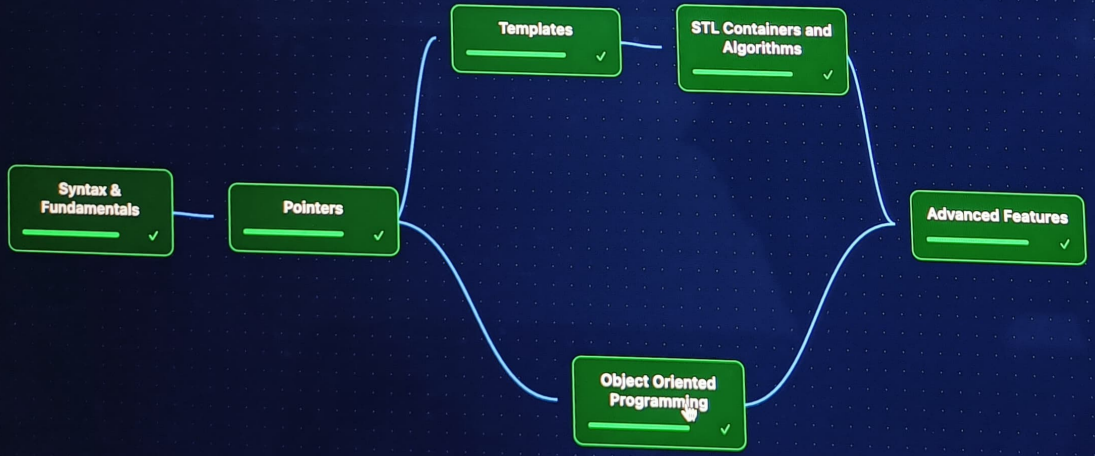
← Roadmap

- Auxiliary Data Types — Easy
- std::array: Fixed-Size Array Cont... — Easy
- std::vector — Medium
- std::list — Medium
- Hashing Containers — Medium
- Ordered Containers — Medium
- std::deque — Medium
- Adapter Containers — Medium

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33/33 topics completed
Expected level



Topics

← Roadmap

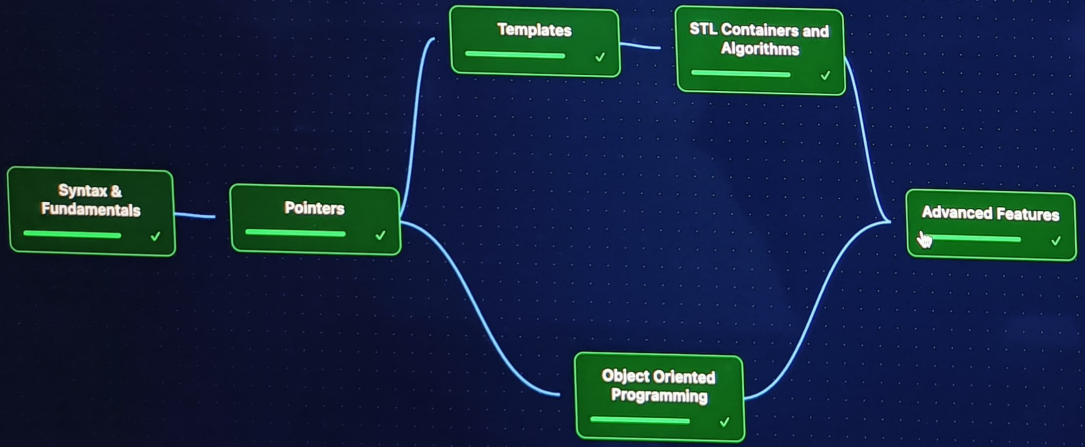
- Classes and Structs — Easy
- Constructor Initialization Lists — Easy
- Inheritance Basics — Easy
- Advanced Constructors — Medium
- Operator Overloading — Medium

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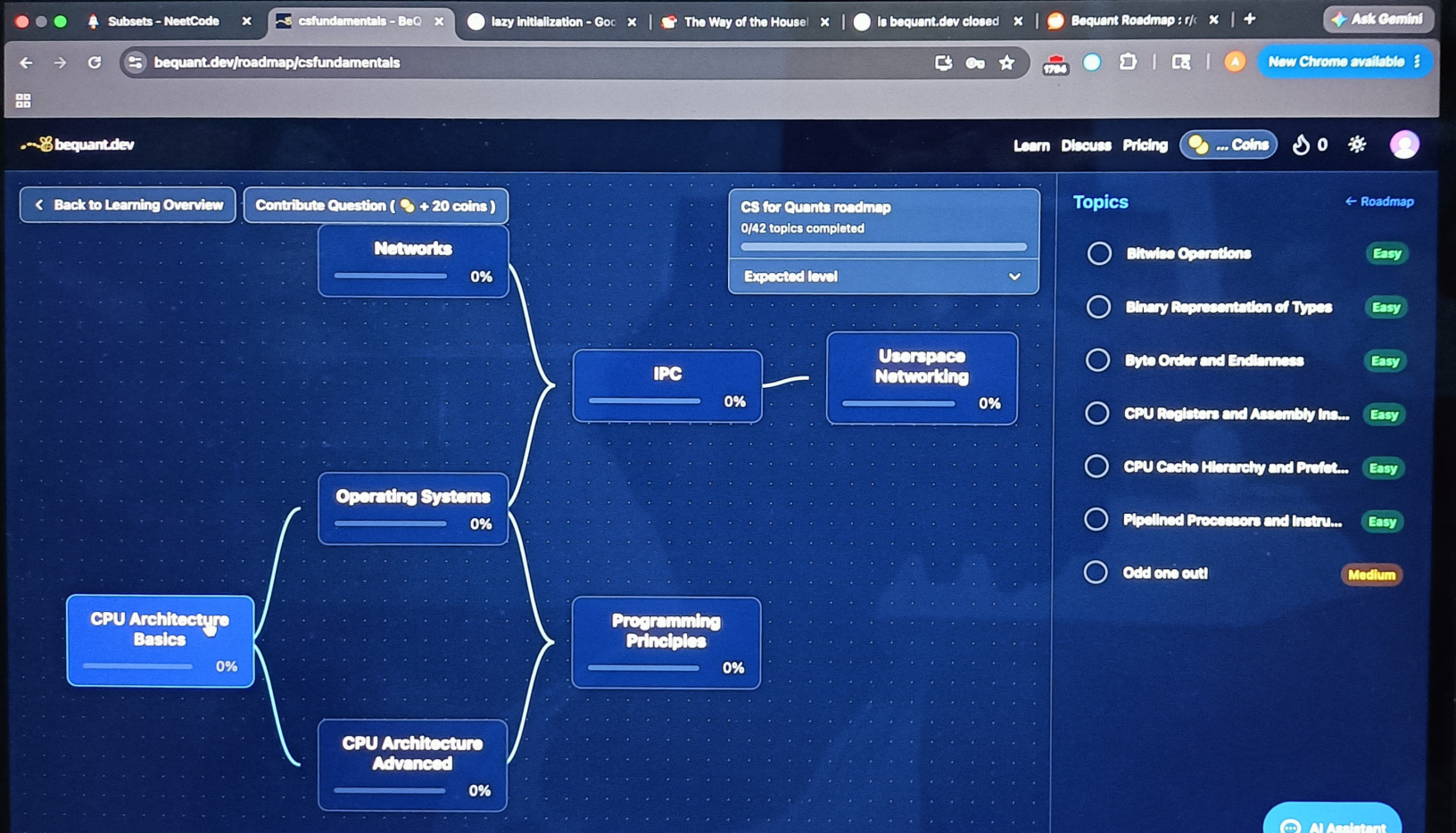
Expected level

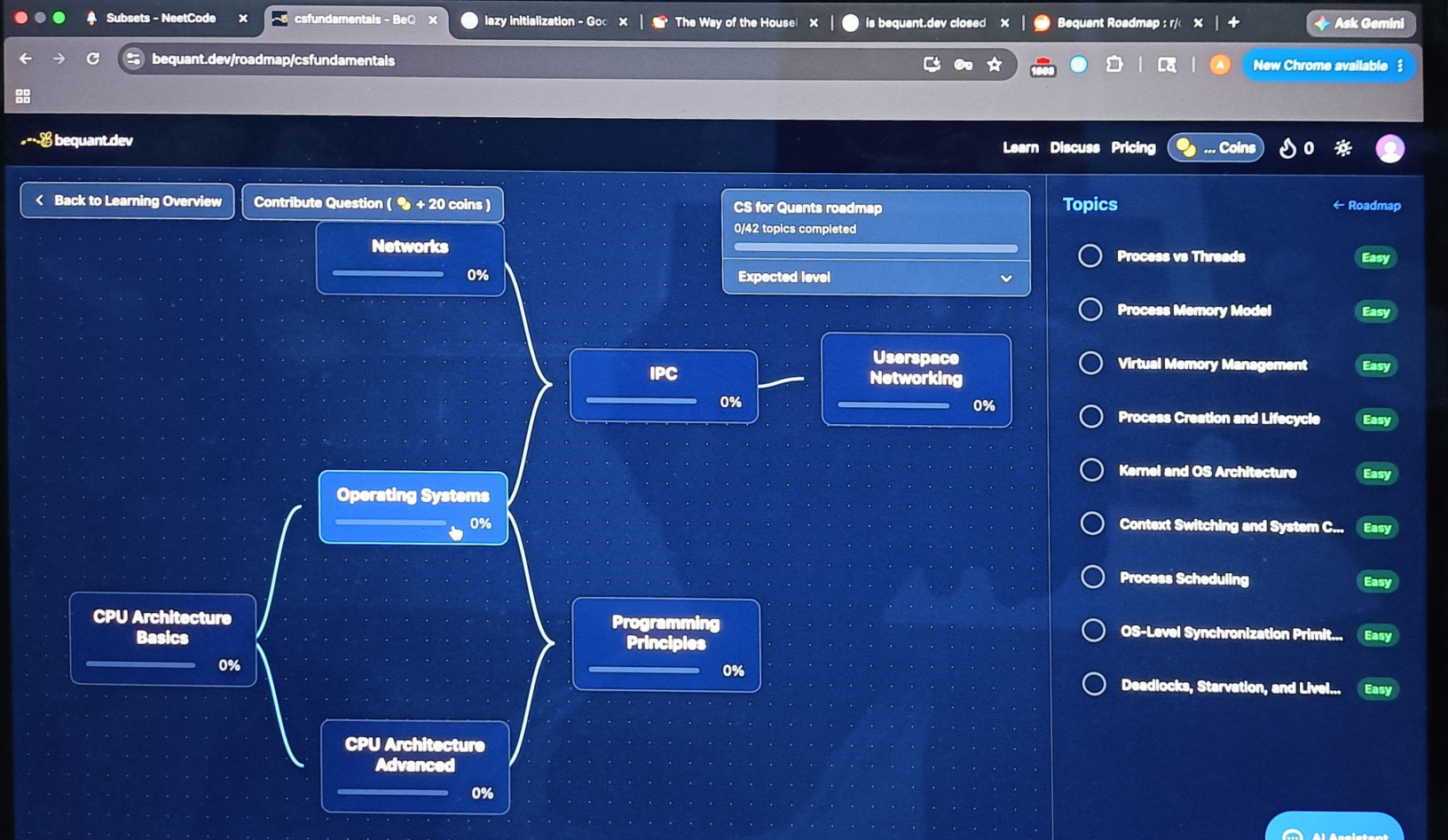


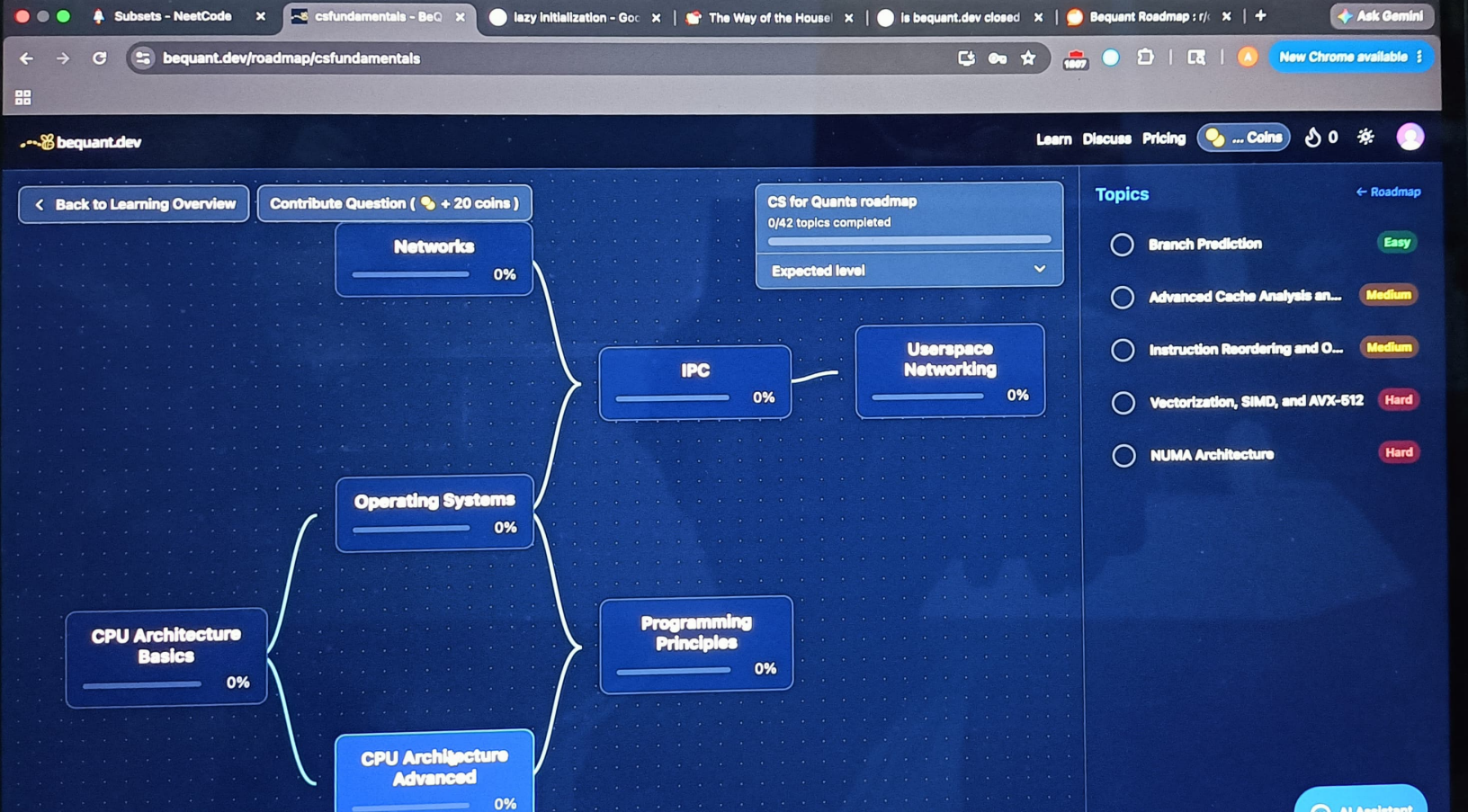
Topics

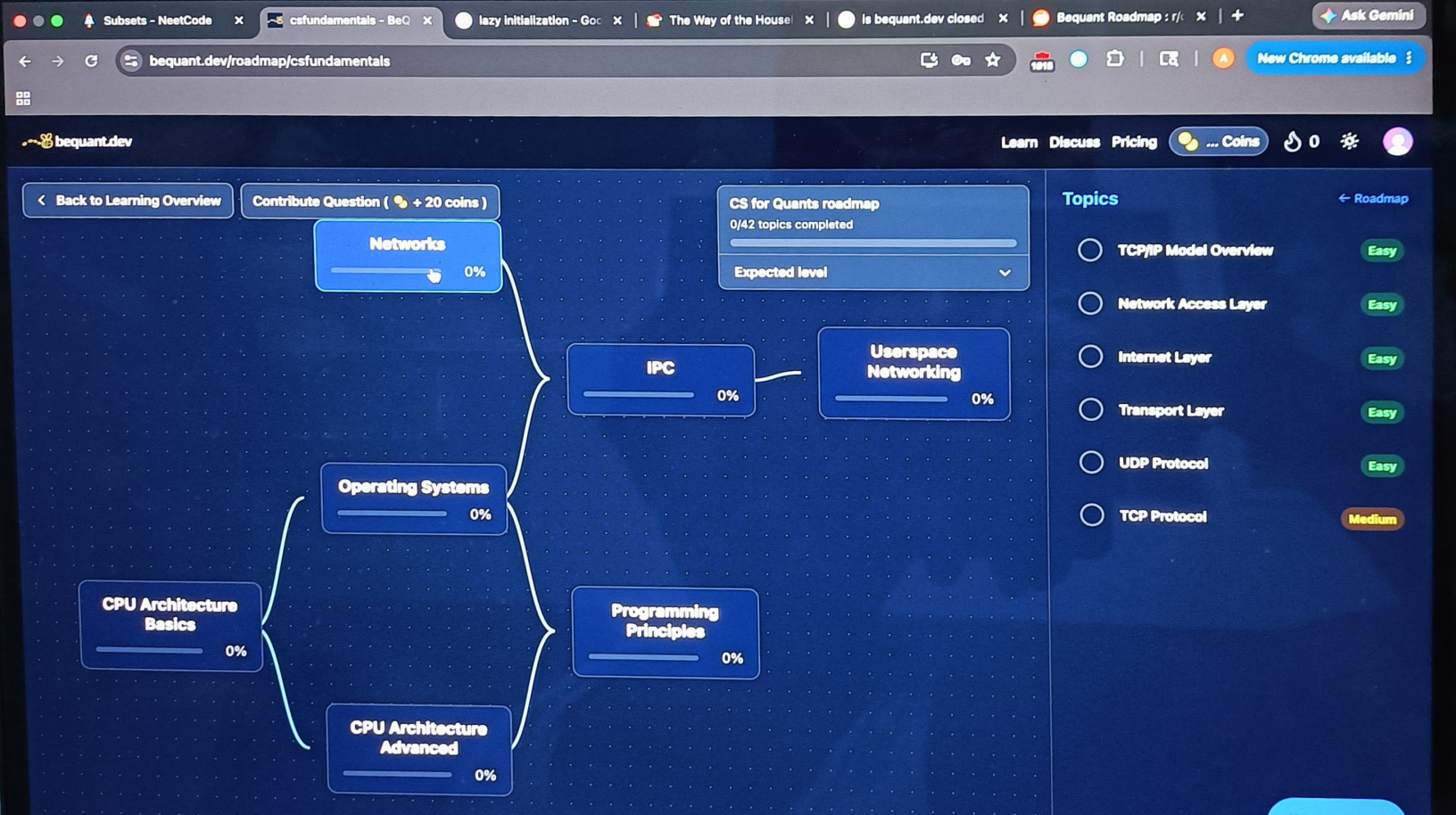
← Roadmap

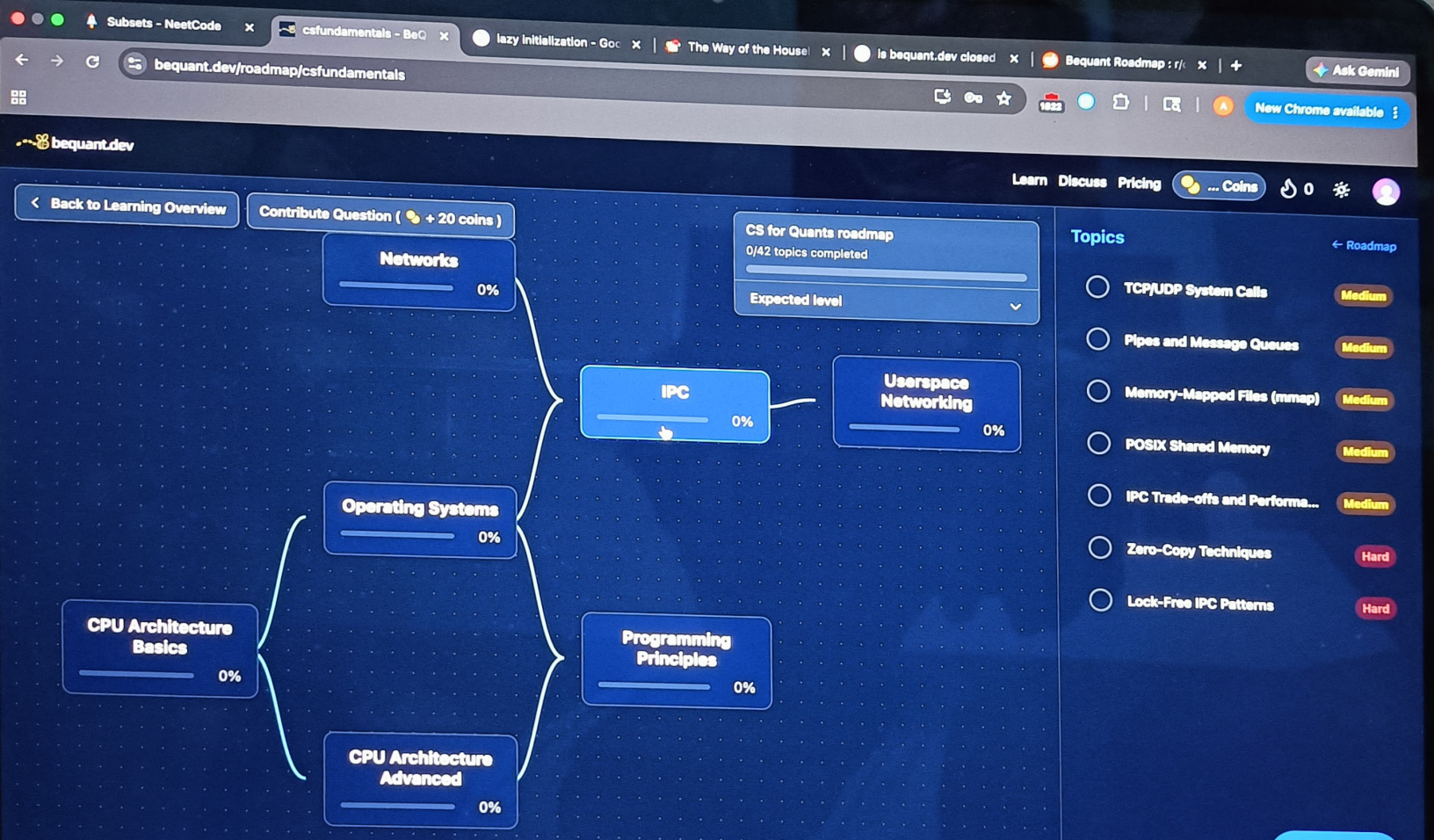
- Type Aliasing — Medium
- Exception Handling — Medium
- Dynamic Casting — Medium
- RTTI, std::any, and std::variant — Hard

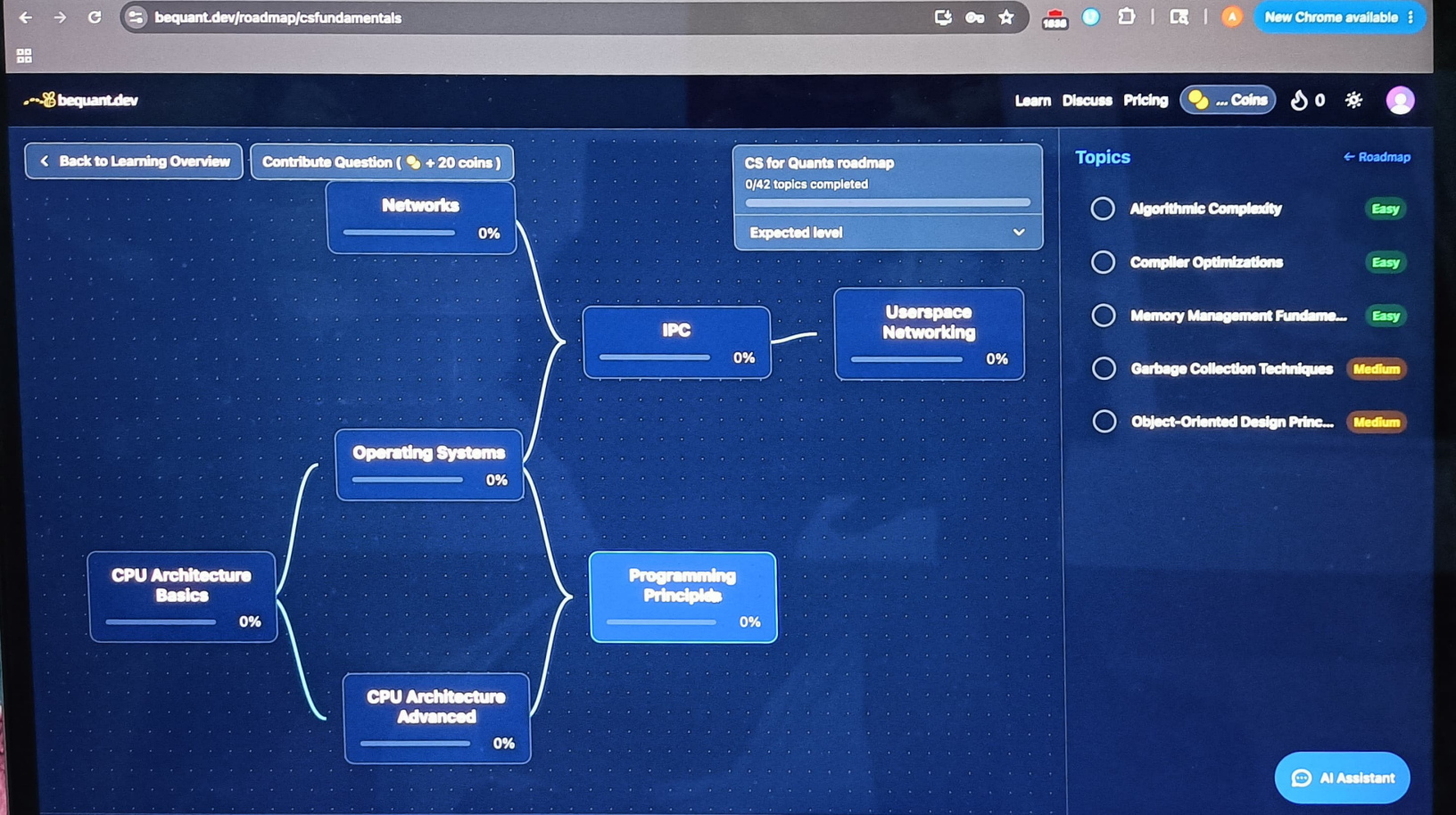

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Expected level
Networks
0%
IPC
0%
Userspace Networking
0%
Operating Systems
0%
CPU Architecture Basics
0%
Programming Principles
0%
CPU Architecture Advanced
0%
Topics
← Roadmap
Bitwise Operations
Easy
Binary Representation of Types
Easy
Byte Order and Endianness
Easy
CPU Registers and Assembly Ins...
Easy
CPU Cache Hierarchy and Prefet...
Easy
Pipelined Processors and Instru...
Easy
Odd one out!
Medium
AI Assistant

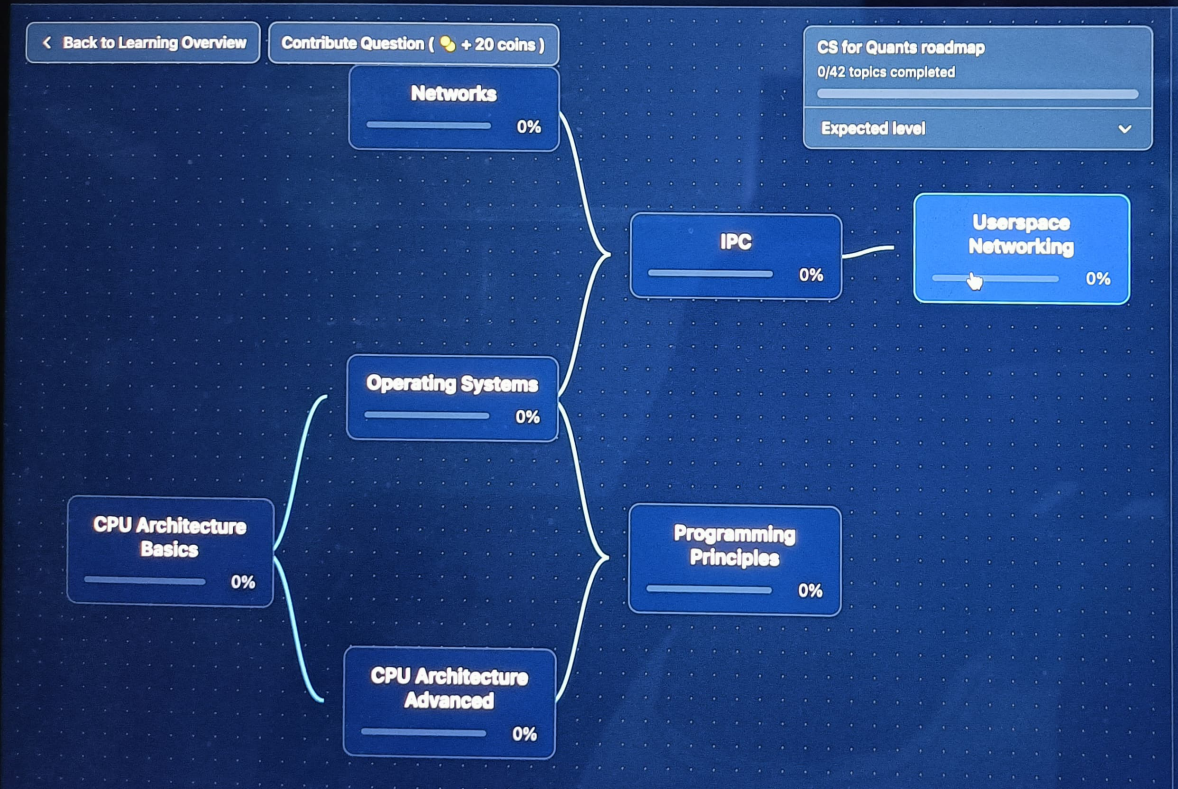

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Networks
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Operating Systems
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CPU Architecture Basics
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Programming Principles
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CPU Architecture Advanced
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Topics
← Roadmap
Process vs Threads
Easy
Process Memory Model
Easy
Virtual Memory Management
Easy
Process Creation and Lifecycle
Easy
Kernel and OS Architecture
Easy
Context Switching and System C...
Easy
Process Scheduling
Easy
OS-Level Synchronization Primit...
Easy
Deadlocks, Starvation, and Livel...
Easy


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Userspace Networking
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Operating Systems
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CPU Architecture Basics
0%
Programming Principles
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CPU Architecture Advanced
0%
Topics
← Roadmap
Branch Prediction
Easy
Advanced Cache Analysis an...
Medium
Instruction Reordering and O...
Medium
Vectorization, SIMD, and AVX-512
Hard
NUMA Architecture
Hard


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Networks
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Userspace Networking
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Operating Systems
0%
CPU Architecture Basics
0%
Programming Principles
0%
CPU Architecture Advanced
0%
Topics
← Roadmap
TCP/IP Model Overview
Easy
Network Access Layer
Easy
Internet Layer
Easy
Transport Layer
Easy
UDP Protocol
Easy
TCP Protocol
Medium


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Networks
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IPC
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Userspace Networking
0%
Operating Systems
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CPU Architecture Basics
0%
Programming Principles
0%
CPU Architecture Advanced
0%
Topics
← Roadmap
TCP/UDP System Calls
Medium
Pipes and Message Queues
Medium
Memory-Mapped Files (mmap)
Medium
POSIX Shared Memory
Medium
IPC Trade-offs and Performa...
Medium
Zero-Copy Techniques
Hard
Lock-Free IPC Patterns
Hard

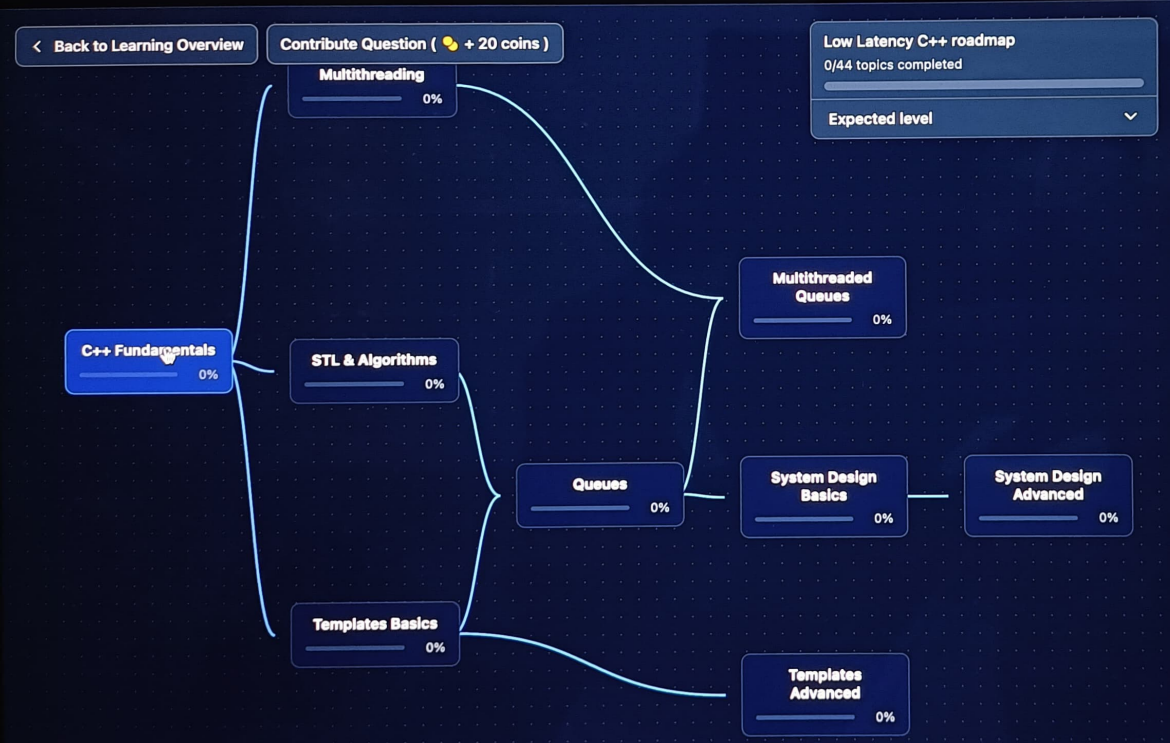

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Networks
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IPC
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Userspace Networking
0%
Operating Systems
0%
CPU Architecture Basics
0%
Programming Principles
0%
CPU Architecture Advanced
0%
Topics
← Roadmap
Algorithmic Complexity
Easy
Compiler Optimizations
Easy
Memory Management Fundame...
Easy
Garbage Collection Techniques
Medium
Object-Oriented Design Princ...
Medium
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Topics

← Roadmap

- Zero-Copy Networking — Medium
- Polling vs Interrupts — Medium
- Kernel Bypass Techniques: DPD... — Hard

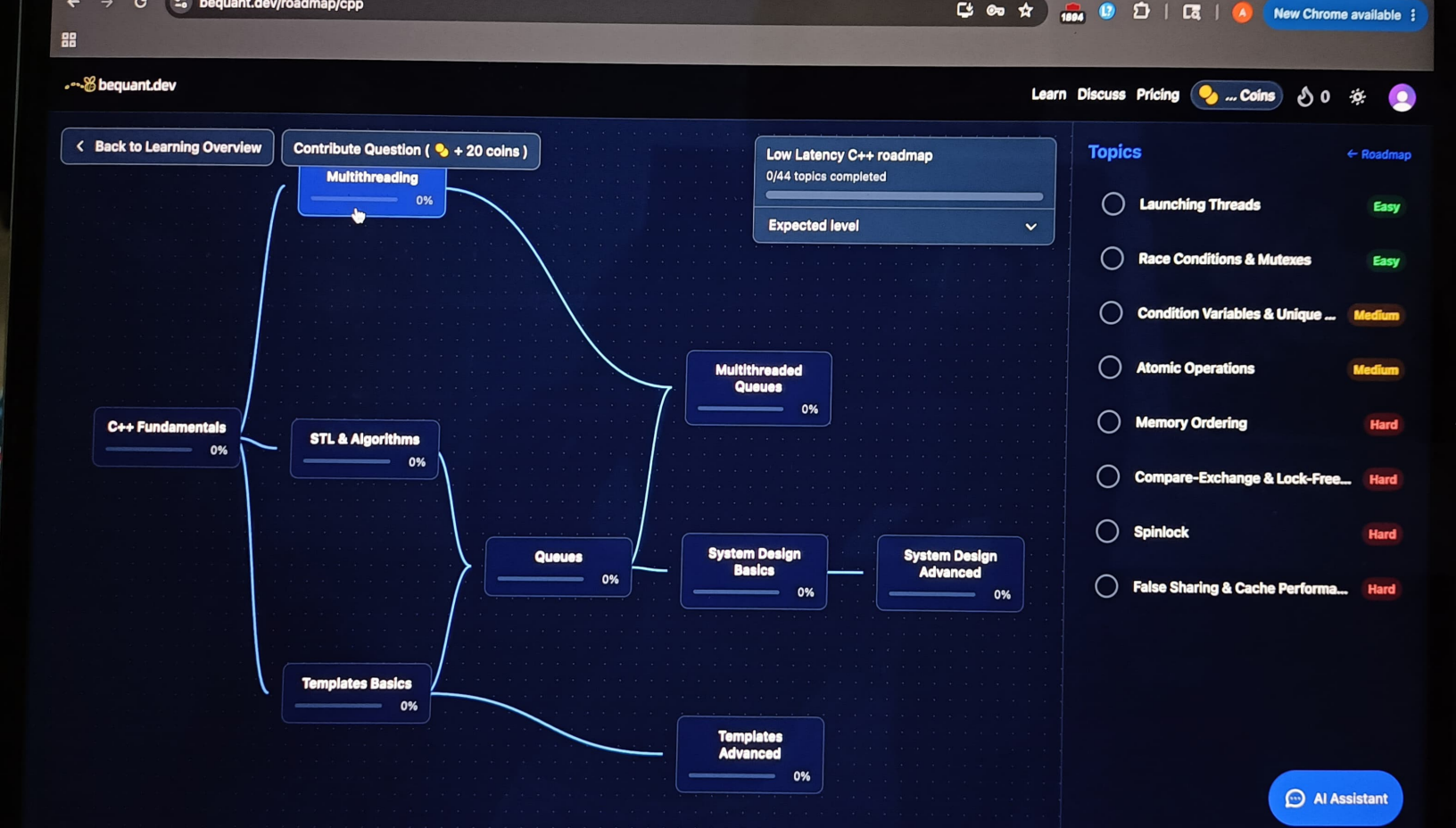


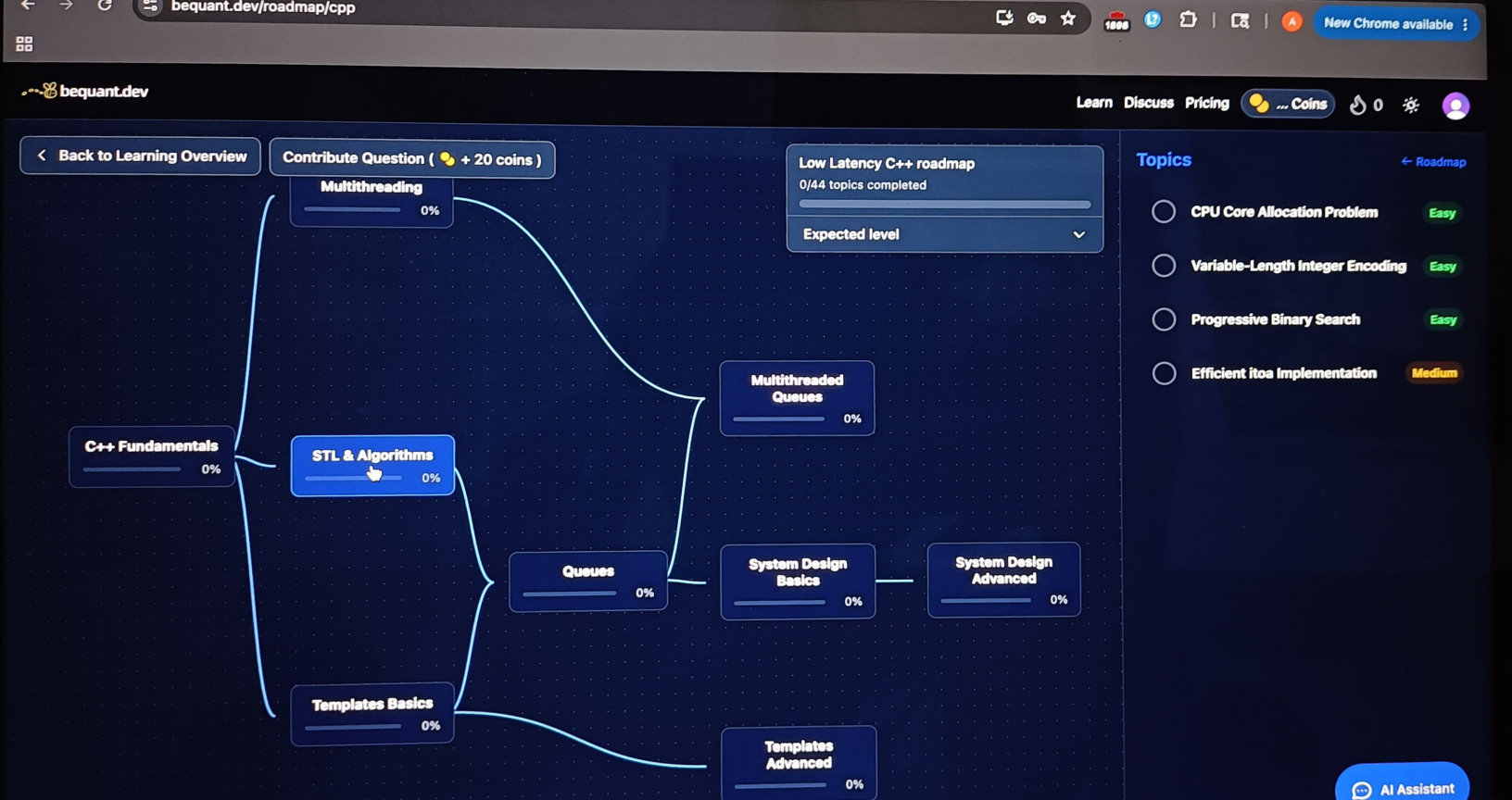
Topics

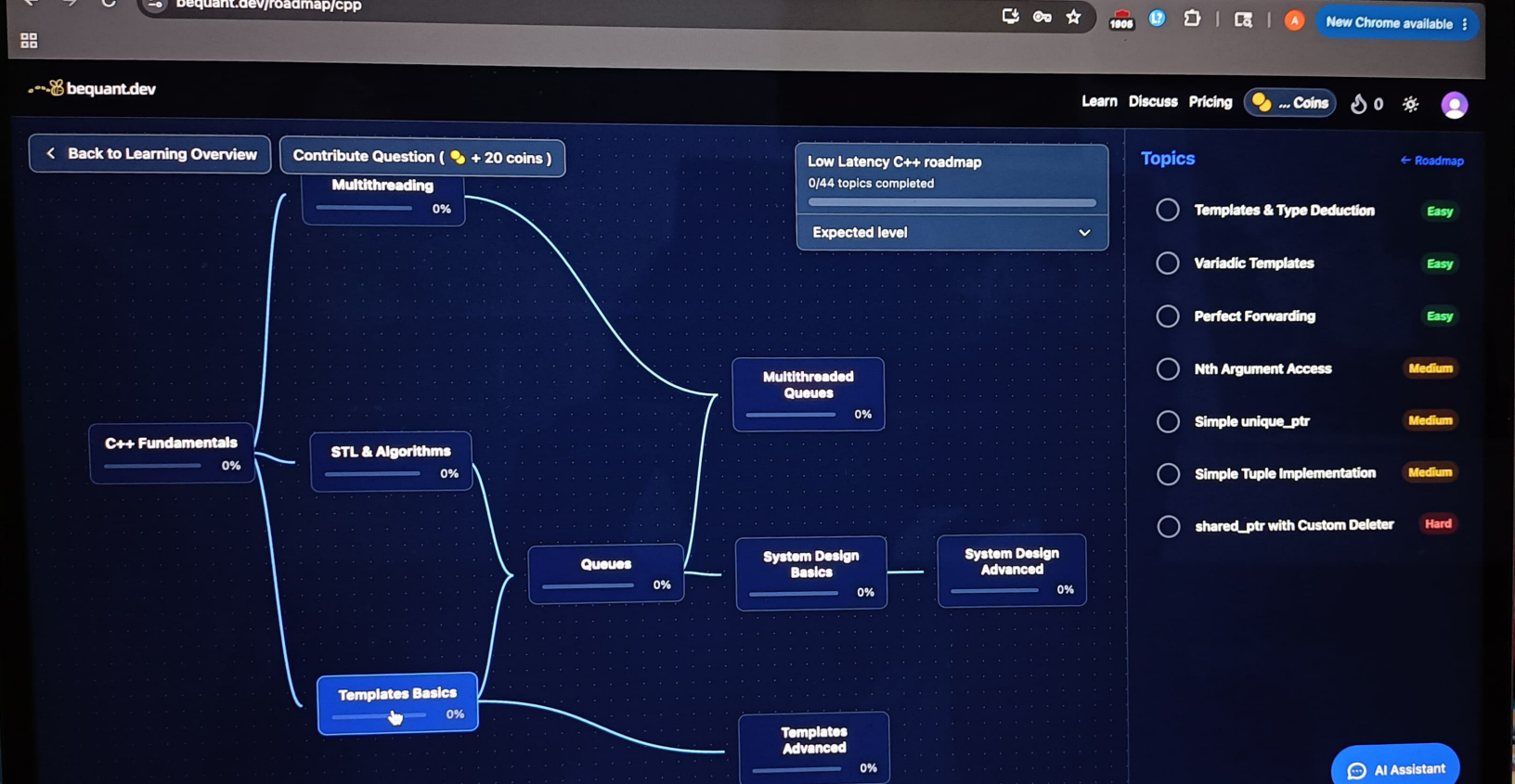
← Roadmap

- Virtual Functions and Inheritance — Easy
- L-values and R-values — Easy
- Copy-Move Elision — Easy
- RAII — Easy
- Smart Pointers — Easy
- Rule of Five — Easy

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Multithreading
0%
C++ Fundamentals
0%
STL & Algorithms
0%
Templates Basics
0%
Queues
0%
Multithreaded Queues
0%
System Design Basics
0%
System Design Advanced
0%
Templates Advanced
0%
Topics
Roadmap
Launching Threads
Easy
Race Conditions & Mutexes
Easy
Condition Variables & Unique ...
Medium
Atomic Operations
Medium
Memory Ordering
Hard
Compare-Exchange & Lock-Free...
Hard
Spinlock
Hard
False Sharing & Cache Performa...
Hard
AI Assistant


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Expected level
Topics
← Roadmap
CPU Core Allocation Problem
Easy
Variable-Length Integer Encoding
Easy
Progressive Binary Search
Easy
Efficient itoa Implementation
Medium
Multithreading
0%
C++ Fundamentals
0%
STL & Algorithms
0%
Templates Basics
0%
Queues
0%
Multithreaded Queues
0%
System Design Basics
0%
System Design Advanced
0%
Templates Advanced
0%
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Expected level
Topics
← Roadmap
Templates & Type Deduction
Easy
Variadic Templates
Easy
Perfect Forwarding
Easy
Nth Argument Access
Medium
Simple unique_ptr
Medium
Simple Tuple Implementation
Medium
shared_ptr with Custom Deleter
Hard
Multithreading
0%
C++ Fundamentals
0%
STL & Algorithms
0%
Multithreaded Queues
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Queues
0%
System Design Basics
0%
System Design Advanced
0%
Templates Basics
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Templates Advanced
0%
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0/44 topics completed

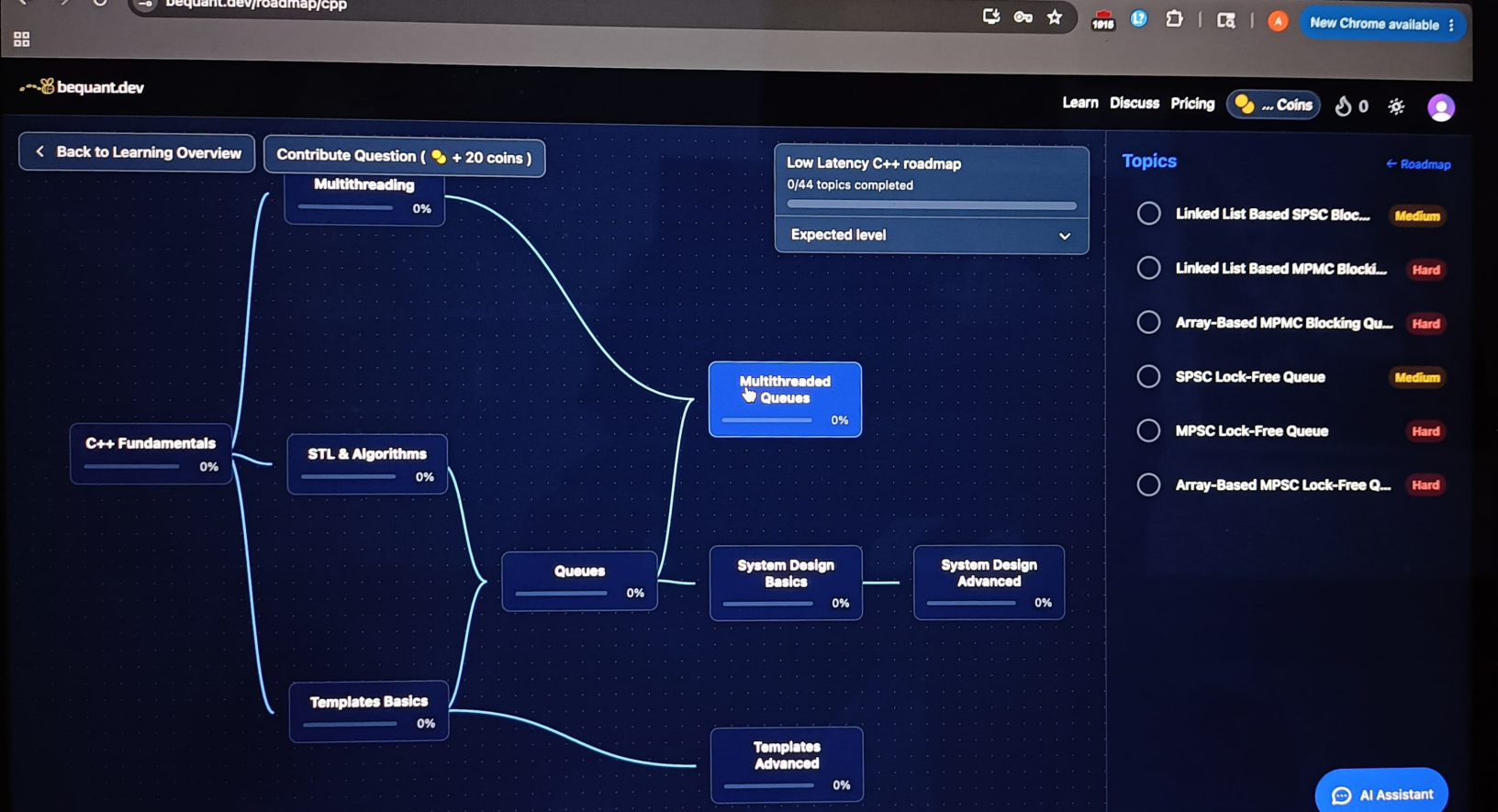
Expected level

Topics

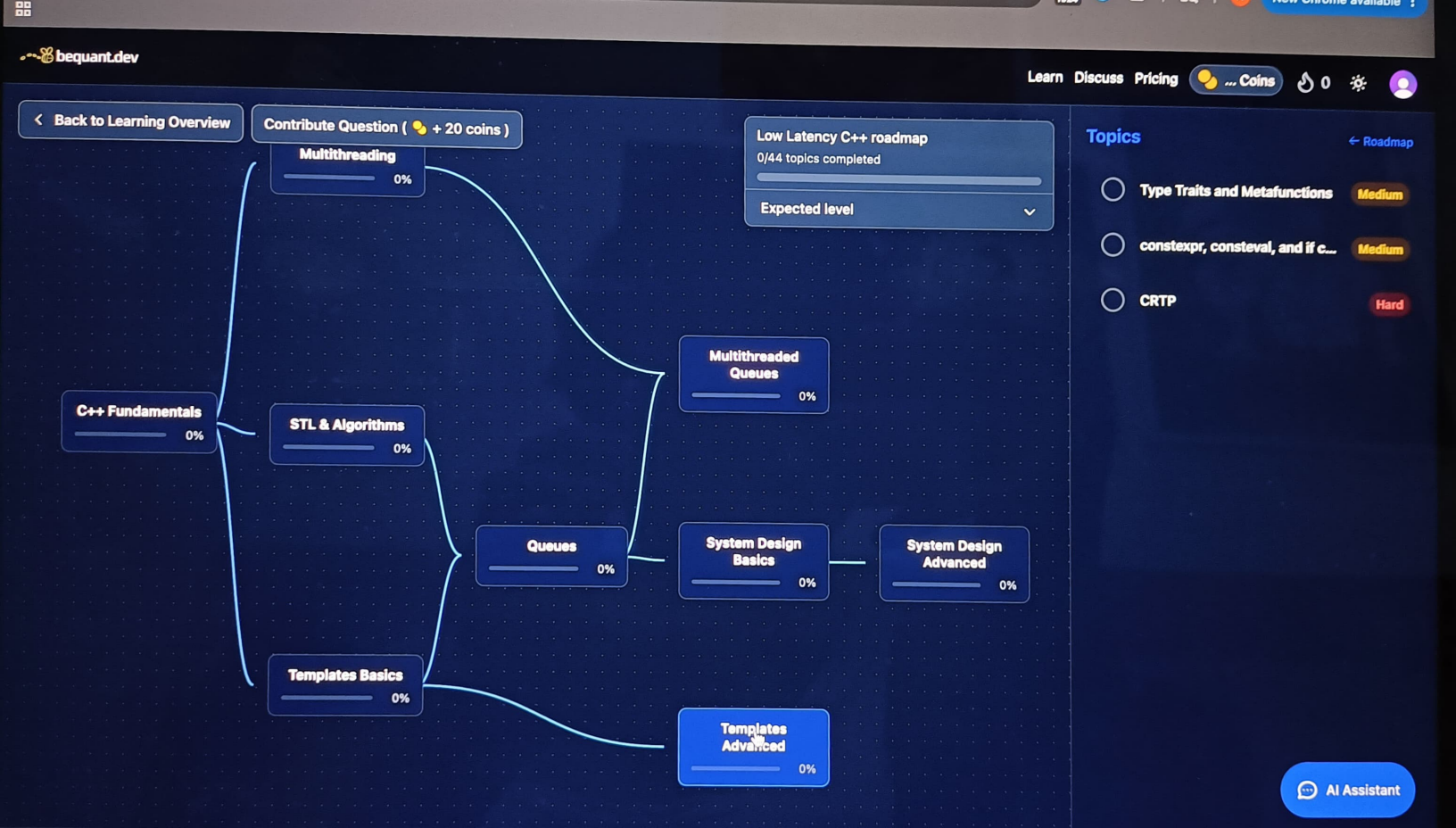
← Roadmap

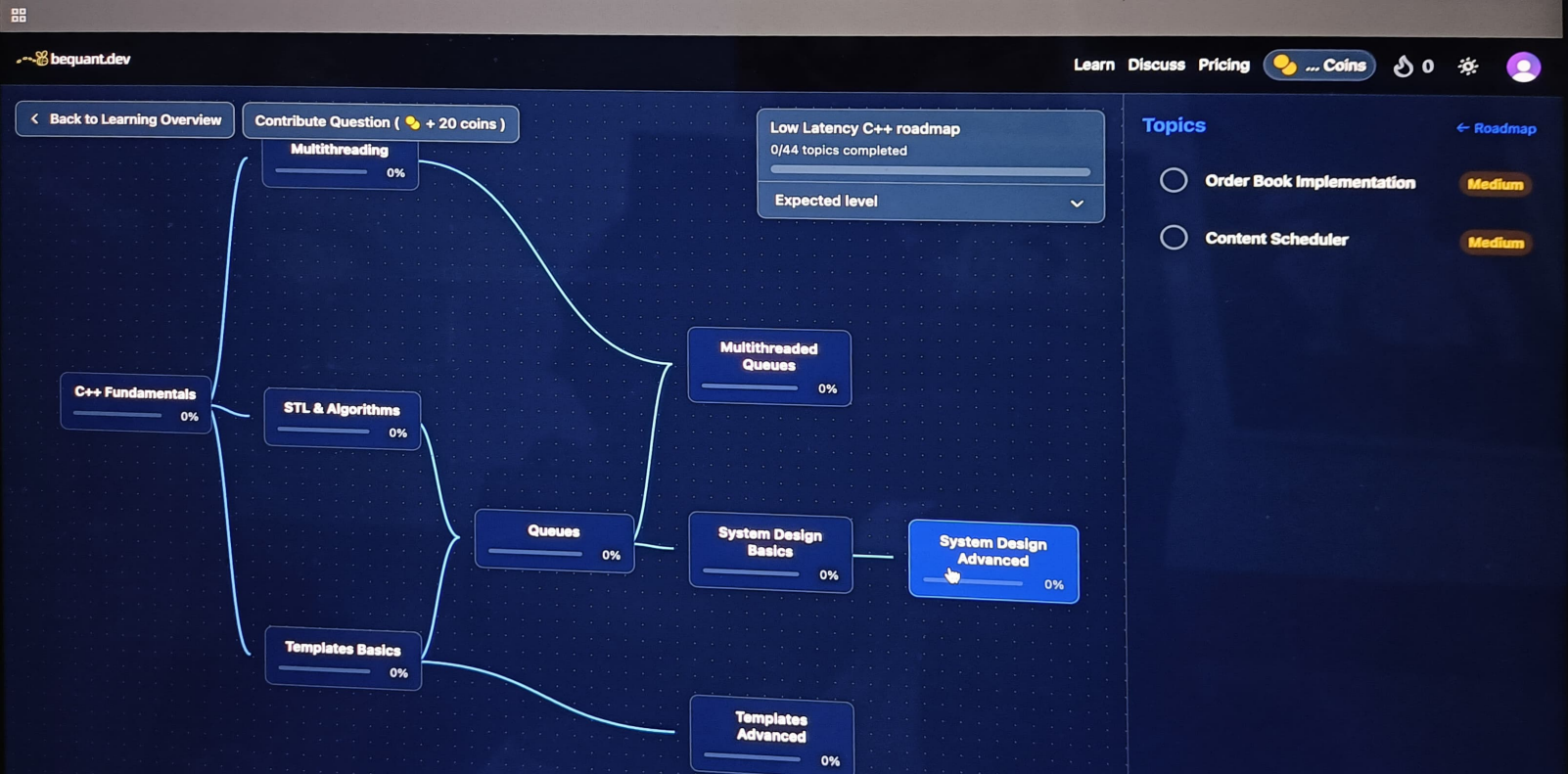
- Linked List based Queue — Easy
- Memory-Pooled Linked List Q... — Medium
- Array based Queue — Medium
- Resizing Array based Queue — Medium
- Segmented Deque — Hard

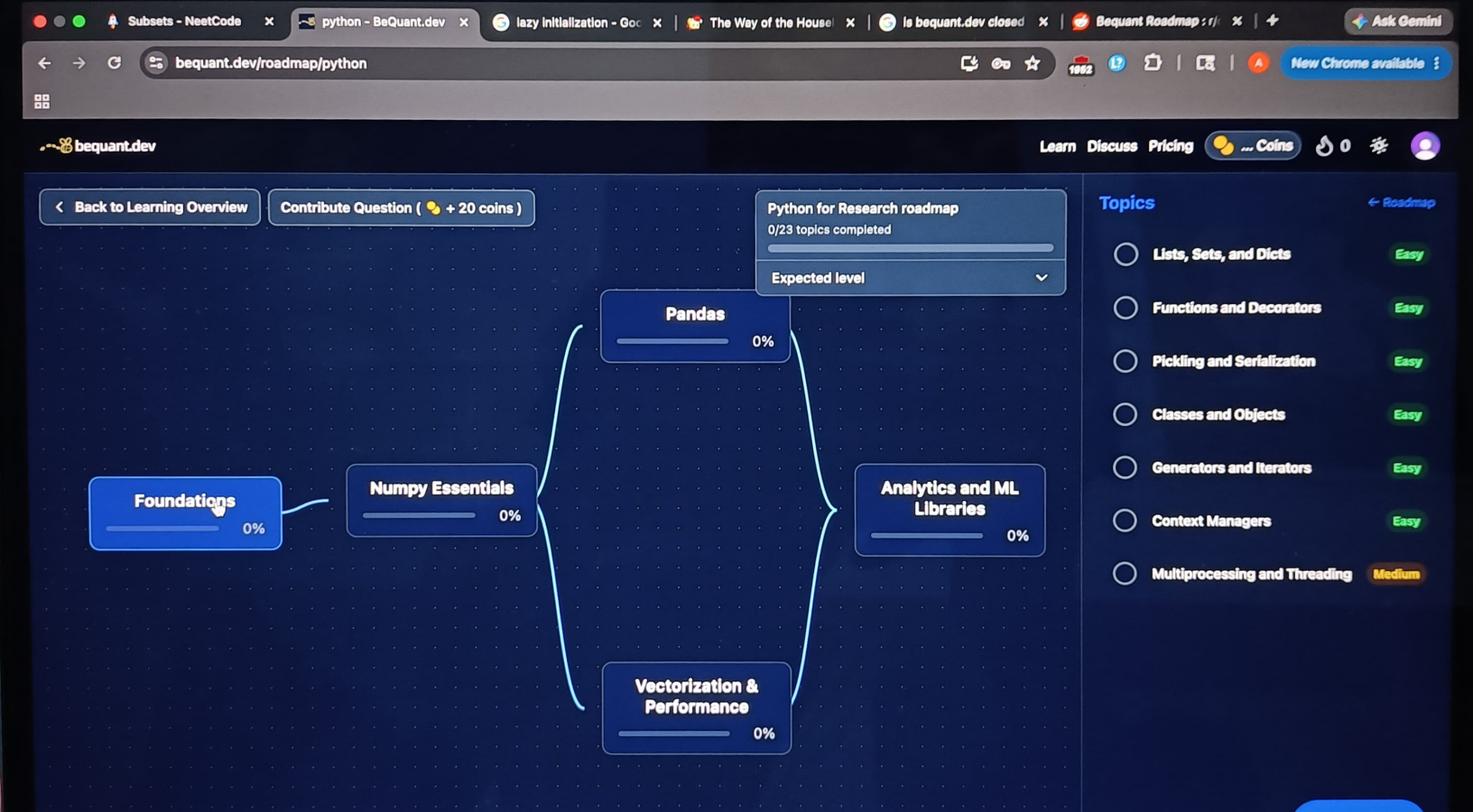
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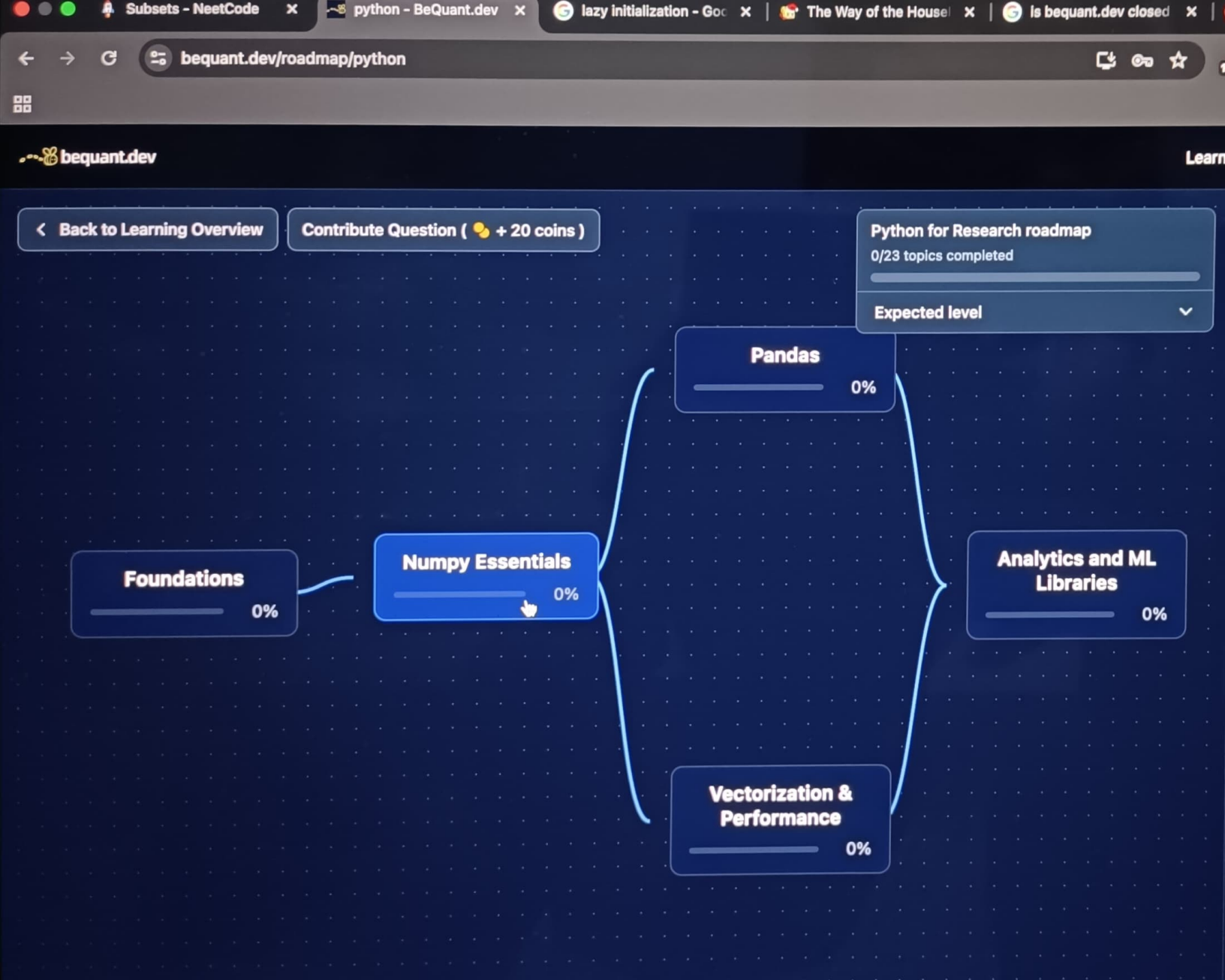

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Expected level
Topics
Roadmap
Multithreading
0%
C++ Fundamentals
0%
STL & Algorithms
0%
Templates Basics
0%
Queues
0%
Multithreaded Queues
0%
System Design Basics
0%
System Design Advanced
0%
Templates Advanced
0%
Linked List Based SPSC Bloc...
Medium
Linked List Based MPMC Blocki...
Hard
Array-Based MPMC Blocking Qu...
Hard
SPSC Lock-Free Queue
Medium
MPSC Lock-Free Queue
Hard
Array-Based MPSC Lock-Free Q...
Hard
AI Assistant

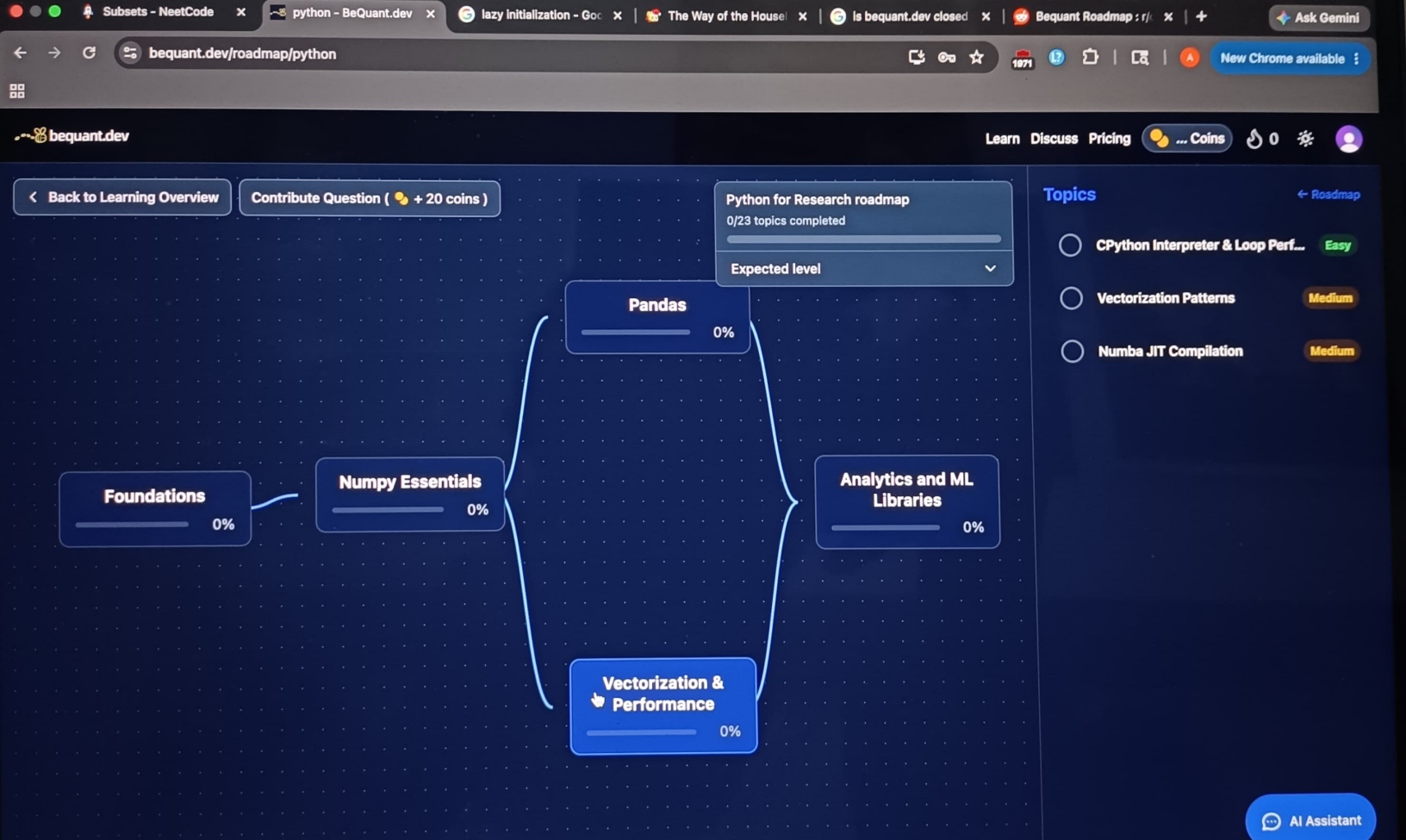

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0/44 topics completed
Expected level
Topics
← Roadmap
Design LRU Cache
Easy
Packed Struct Offset Lookup
Medium
Design LFU Cache
Hard
Multithreading
0%
C++ Fundamentals
0%
STL & Algorithms
0%
Templates Basics
0%
Queues
0%
Multithreaded Queues
0%
System Design Basics
0%
System Design Advanced
0%
Templates Advanced
0%
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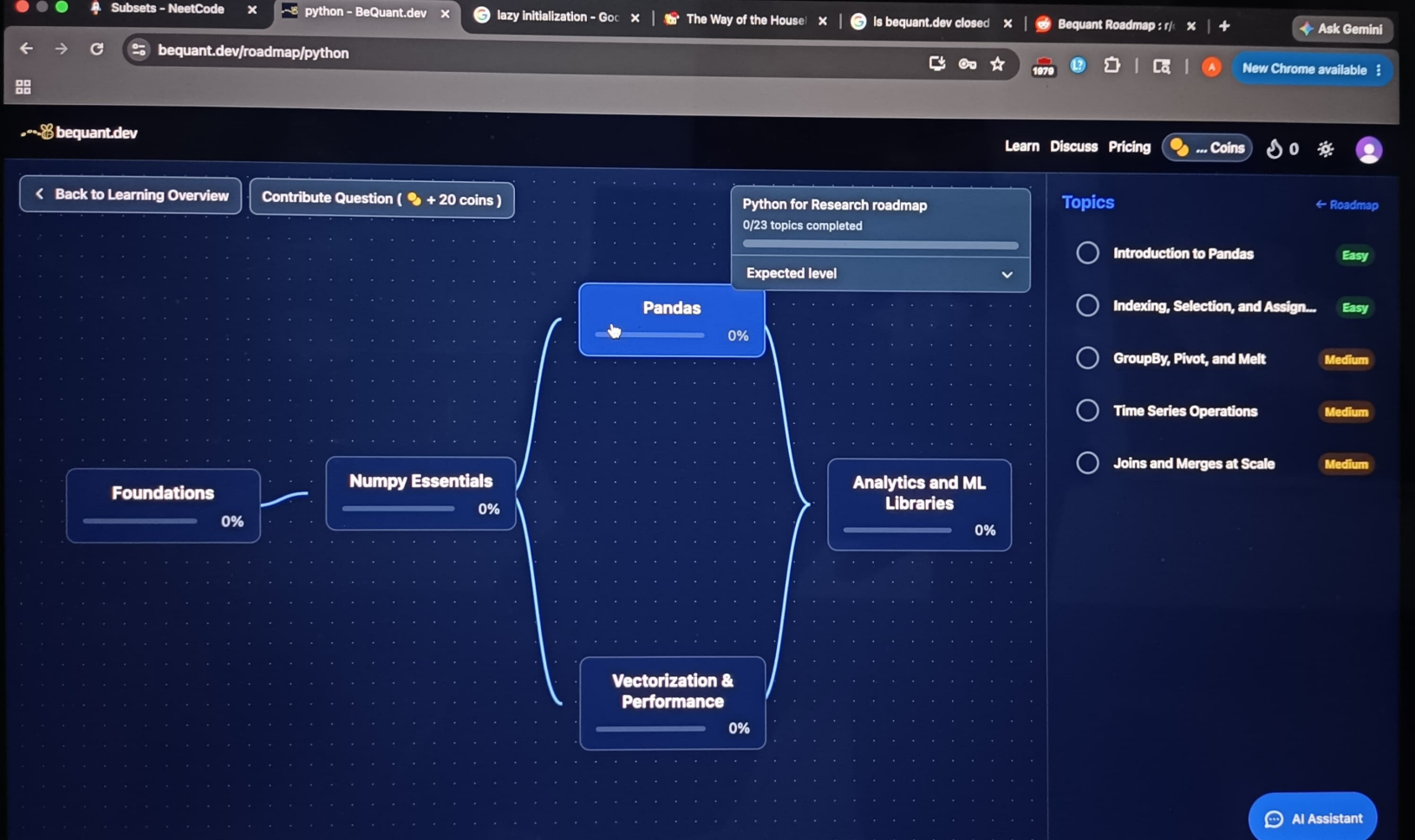

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0/44 topics completed
Expected level
Topics
Roadmap
Type Traits and Metafunctions
Medium
constexpr, consteval, and if c...
Medium
CRTP
Hard
Multithreading
0%
C++ Fundamentals
0%
STL & Algorithms
0%
Templates Basics
0%
Queues
0%
Multithreaded Queues
0%
System Design Basics
0%
System Design Advanced
0%
Templates Advanced
0%
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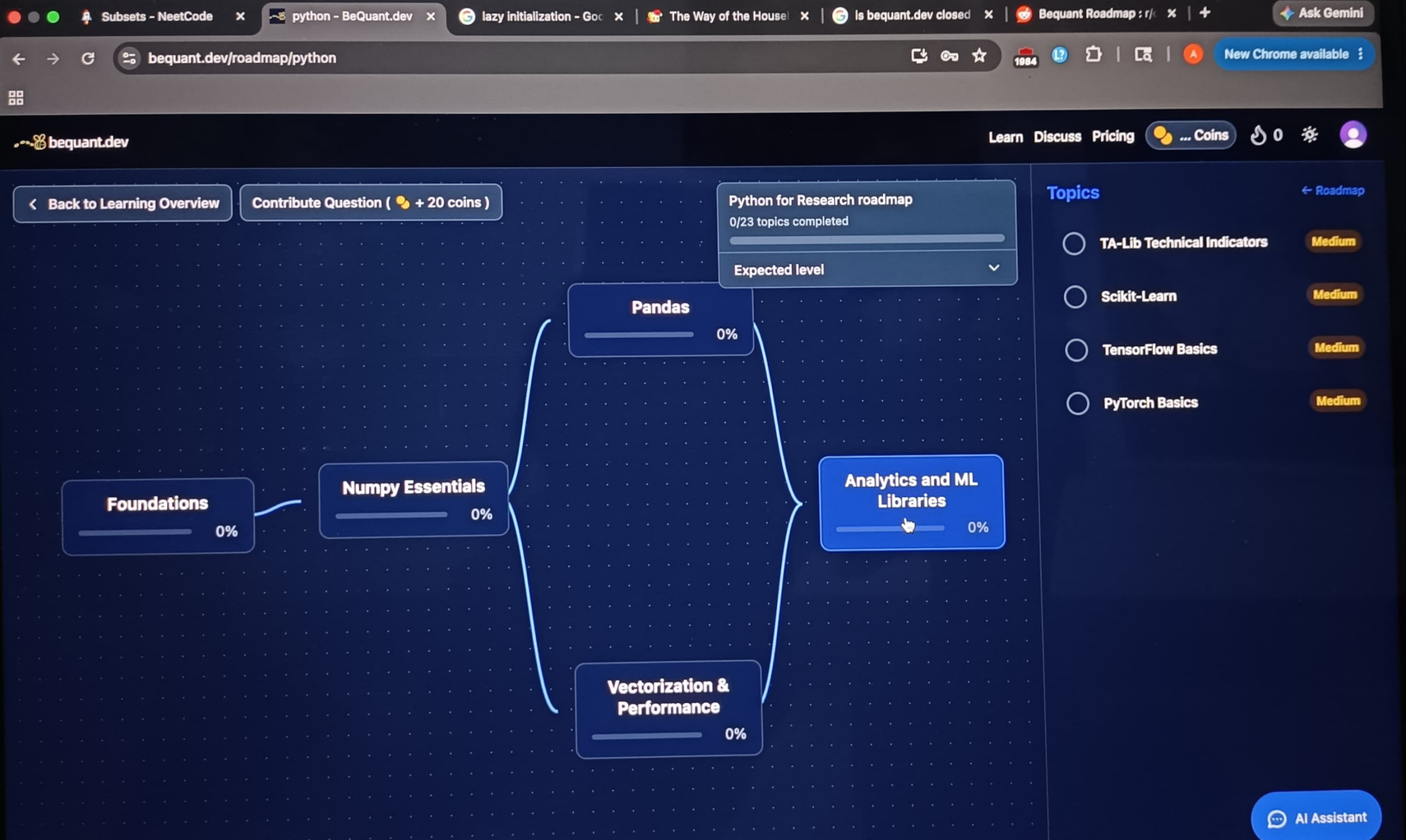

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Expected level
Topics
← Roadmap
Order Book Implementation
Medium
Content Scheduler
Medium
Multithreading
0%
C++ Fundamentals
0%
STL & Algorithms
0%
Templates Basics
0%
Queues
0%
Multithreaded Queues
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System Design Basics
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System Design Advanced
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Templates Advanced
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Python for Research roadmap
0/23 topics completed
Expected level
Pandas
0%
Foundations
0%
Numpy Essentials
0%
Analytics and ML Libraries
0%
Vectorization & Performance
0%
Topics
← Roadmap
Lists, Sets, and Dicts
Easy
Functions and Decorators
Easy
Pickling and Serialization
Easy
Classes and Objects
Easy
Generators and Iterators
Easy
Context Managers
Easy
Multiprocessing and Threading
Medium


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Expected level
Pandas
0%
Foundations
0%
Numpy Essentials
0%
Analytics and ML Libraries
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Vectorization & Performance
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Foundations
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Numpy Essentials
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